

The diagram illustrates the sulfur cycle with the following components and reactions:

- Species:**
 - HS^- (top center)
 - $\text{S}(0)$ (top right)
 - SO_3^{2-} (middle left)
 - $\text{S}_2\text{O}_3^{2-}$ (center)
 - $\text{S}_4\text{O}_6^{2-}$ (middle right)
 - SO_3^{2-} (bottom right)
 - SO_4^{2-} (bottom center)
 - APS** (bottom left)
- Enzymes and Pathways:**
 - Reduction of HS^- to $\text{S}(0)$:** *FccAB/Sqr*
 - Oxidation of $\text{S}(0)$ to HS^- :** *SOR* (dashed), *PsrABC/ShyABCD* (dashed), *SudAB/Npsr/Nsr* (dashed)
 - Reduction of $\text{S}(0)$ to SO_3^{2-} :** *SOR* (dashed), *sHdrAB1B2B3C1C2*, *DsrABCEFH*, *Sdo*
 - Oxidation of SO_3^{2-} to $\text{S}(0)$:** *TetH*, *SoxB*
 - Reduction of SO_3^{2-} to SO_4^{2-} :** *SoeABC*, *SorAB*, *SorT*
 - Oxidation of SO_4^{2-} to $\text{S}_2\text{O}_3^{2-}$:** *Sat*
 - Reduction of SO_4^{2-} to SO_3^{2-} :** *TetH*, *SoxB*
 - Reduction of $\text{S}_2\text{O}_3^{2-}$ to SO_3^{2-} :** *PhsABC*
 - Oxidation of $\text{S}_2\text{O}_3^{2-}$ to HS^- :** *PhsABC*
 - Reduction of $\text{S}_2\text{O}_3^{2-}$ to SO_4^{2-} :** *SoxYZXAB* (dashed), *SoxYZXABCD* (dashed)
 - Oxidation of SO_4^{2-} to $\text{S}_2\text{O}_3^{2-}$:** *SoxYZXAB* (dashed), *SoxYZXABCD* (dashed)
 - Reduction of $\text{S}_2\text{O}_3^{2-}$ to $\text{S}_4\text{O}_6^{2-}$:** *TsdA/DoxDA* (dashed), *Ttr/Otr* (dashed)
 - Oxidation of $\text{S}_4\text{O}_6^{2-}$ to $\text{S}_2\text{O}_3^{2-}$:** *TsdA/DoxDA* (solid), *Ttr/Otr* (solid)
 - Reduction of $\text{S}_4\text{O}_6^{2-}$ to SO_3^{2-} :** *TetH* (dashed), *SoxB* (dashed)
 - Reduction of $\text{S}_4\text{O}_6^{2-}$ to SO_4^{2-} :** *TetH* (solid), *SoxB* (solid)
 - Reduction of SO_3^{2-} to **APS**:** *AprAB* (green), *QmoABC* (boxed)
 - Oxidation of **APS** to SO_3^{2-} :** *DsrMKJOP* (boxed), *e^-*
 - Reduction of HS^- to **APS**:** *DsrMKJOP* (boxed), *e^-*

[illegible]

The diagram illustrates the dissimilatory sulfide oxidation pathway in *Shewanella putrefaciens*, showing the flow of electrons from sulfide through various proteins across the cell membrane.

Periplasm:

- FccAB** and **Sqr** are involved in the oxidation of HS^- to HS_n^- .
- Sqr** is coupled with the reduction of menaquinone ($\text{MQ} \rightarrow \text{MQH}_2$).
- S⁰** is shown as a starting point for the oxidation of R-SH to R-S-SH .

Membrane:

- DsrMKJOP** is involved in the reduction of menaquinone ($\text{MQ} \rightarrow \text{MQH}_2$).
- QmoABC** is involved in the reduction of menaquinone ($\text{MQ} \rightarrow \text{MQH}_2$).

Cytoplasm:

- DsrEFH** is involved in the oxidation of HS-SH to S-SH .
- DsrC** is a central protein that can exist as a monomer or a dimer (S-S(0)).
- DsrBA** and **DsrL** are involved in the oxidation of S-SH to SO_3^{2-} .
- AprBA** is involved in the oxidation of SO_3^{2-} to APS .
- Sat** is involved in the oxidation of APS to SO_4^{2-} .

Electron Flow:

- Electrons from HS^- and S^0 are transferred through **Sqr** and **DsrMKJOP** to the membrane.
- Electrons from R-SH are transferred through **DsrEFH** and **DsrC** to the membrane.
- Electrons from SO_3^{2-} are transferred through **AprBA** and **Sat** to the membrane.

Legend:

- Orange:** FccAB, Sqr, DsrEFH
- Green:** MQ $\rightarrow$ MQH₂, AprBA
- Blue:** DsrC
- Yellow:** DsrBA, DsrL
- Red:** S⁰, S-S(0), SO₃²⁻, APS, SO₄²⁻

Dissimilatory sulfide oxidation pathway